MILITARY SPECIFICATION

SYRINGE, CARTRIDGE, ASPIRATING, THUMB RING HANDLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an aspirating-type syringe suitable for use with local anesthetic solutions in glass cartridges of 1.8 ml capacity.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GG-I-526	-	Instruments, Dental and Surgical, General Specifications.
QQ-N-290	-	Nickel Plating (Electrodeposited)
QQ-C-320	-	Chromium Plating (Electrodeposited)
MMM-A-250	-	Adhesive, Water-Resistant, (For sealing Fiberboard Boxes).
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-585	-	Box, Wood, Wirebound.
PPP-B-601	-	Box, Wood, Cleated-Plywood.
PPP-B-621	-	Box, Wood, Nailed and Lock Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Box, Set-Up.

Beneficial comments (recommendatirs, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, Pa. 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116 - Preservation-packaging, Methods of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)

ASTM A380 - Cleaning and Descaling Stainless Steel Parts, Equipment, and Systems.
ASTM E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
ASTM E92 - Vickers Hardness of Metallic Materials.
ASTM E140 - Standard Hardness Conversion Tables for Metals.

3. REQUIREMENTS

3.1 General requirements. The aspirating syringe shall be in accordance with the applicable requirements of GG-I-526 and as specified herein.

3.2 Material. The harpoon and piston assembly shall be fabricated from class 3, type 302, type 303 or type 304 stainless steel in accordance with GG-I-526. All other components shall be fabricated from class 3 material as stated above or a suitable copper-zinc alloy.

3.2.1 Hardness. After proper stress relieving all components fabricated of class 3 material shall exhibit a Rockwell hardness of B90 to B100 when tested as specified in 4.4.1.

3.3 Style, design and construction. The syringe shall be of the aspirating type designed for use with disposable needles. The syringe shall be of the side loading type for use with 1.8 milliliter (ml) cartridges having an outside diameter of 8.25 to 9.1 millimeters (mm) and an overall length of 63 to 63.5 mm. The syringe shall consist of a barrel, thumb ring, finger

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grip, spring, guide bearing, plunger with harpoon and an adapter. The adapter shall be furnished with a male thread for securing the disposable needles and a 0.028 to 0.032 inch diameter hole at the center to accommodate a 25 gage cartridge needle. The harpoon shall be suitable for engaging the rubber piston of a glass anesthetic solution cartridge of the 1.8 ml capacity. The entire syringe shall be autoclavable. The dimensions and tolerances shall be as specified in figure 1.

3.4 Finish. All components fabricated from class 3 materials shall be passivated in accordance with ASTM A380.

3.5 Plating. All components fabricated from copper-zinc alloy shall be chromium plate over nickel plate. The nickel plate shall be in accordance with QQ-N-290, class 1, grade E. The chromium plate shall be in accordance with QQ-C-320, class 1, type 1.

3.6 Salt spray. The syringe shall be capable of withstanding the salt spray test specified in GG-I-526.

3.7 Boil test. The syringe shall be capable of withstanding the boil test specified in GG-I-526.

3.8 Performance. The harpoon shall not become disengaged from the cartridge piston when tested as specified in 4.4.2.

3.9 Instructions. Each unit package shall contain an insert providing adequate instructions for use of the syringe. Instructions shall include information for proper loading of cartridge and attachment of the needle.

3.10 Government furnished material. The government shall furnish the contractor the cartridges required to perform the piercing test specified in 4.5.1.

3.11 Identification marking. The barrel of each syringe shall be permanently and legibly marked, by engraving or embossing, with the registered trademark or name of the manufacturer.

3.12 Spare parts. A spare plunger with harpoon shall be supplied with each syringe.

3.13 Workmanship. Workmanship shall be first class throughout. Item shall be free from defects which detract from its appearance or may impair its serviceability. All burrs and sharp edges shall be removed from the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the

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contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I. The unit of product for examination purposes shall be one syringe.

TABLE I. Sampling for examination.

Type of defect	Inspection level	AQL (percent defective)
Major	I	1.0
Minor	I	4.0
Dimensional	S-1	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table II. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics. The unit of product for test purposes shall be one syringe.

TABLE II. Sampling for tests.

Component	Characteristic	Requirement	Test Procedure	Inspection level	AQL (percent defective)
Syringe	Hardness	3.2.1	4.4.1	S-1	2.5
	Thickness of plating	3.5	QQ-N-290, QQ-C-320	S-1	2.5
	Salt spray	3.6	GG-I-526	S-1	2.5
	Boil test	3.7	GG-I-526	S-1	2.5
	Performance	3.8	4.4.2	1/	1/

1/ Twelve (12) inspected and approved cartridges shall be tested with twelve (12) different syringes. Failure of any syringe tested shall be cause for rejection of the lot represented.

4.2.3 Raw material. A sufficient quantity of each lot of metal used in the manufacture of the syringe shall be selected for composition analysis in accordance with GG-I-526.

4.3 Examination.

4.3.1 Syringe. The syringe shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination however, shall not be restricted to defects listed in table III. Any syringe in the sample containing one or more defects shall be rejected and if the number of defective syringes in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

TABLE III. Classification of defects.

Categories	Defects
Major:	
101	Plunger not retractable.
102	Barrel does not accommodate specified cartridge.
103	Orifice in adapter not as specified.
104	Instrument not free of burrs or sharp edges.
105	Threads at distal end not cut cleanly.
Minor:	
201	Plating non-adherent.
202	Plunger action too loose radially permitting plunger to assume askew-position.

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4.3.2 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Hardness test. Hardness shall be determined in accordance with ASTM E18 or ASTM E92. Conversion of Vickers hardness values into Rockwell values shall be in accordance with ASTM E140.

4.4.2 Performance test. A cartridge shall be inserted in the syringe, with attached needle, per manufacturer's instructions. The rubber plunger of the cartridge shall be pierced by the harpoon and the plunger pushed forward, with moderate presssure, for a distance of 1/2 inch. The plunger shall then be pulled in the opposite direction, as in aspiration.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Unit of issue. One syringe along with instructions for use and one replacement plunger with harpoon, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Cleaning. Syringes and replacment plungers with harpoons shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.2.2 Drying. Syringes and replacement plungers with harpoons shall be dried in accordance with MIL-P-116.

5.1.2.3 Preservative application. Preservatives shall not be used.

5.1.2.4 Unit packaging. Each syringe along with replacement plunger and harpoon assembly shall be packaged in accordance with MIL-P-116, method III within a box conforming to PPP-B-676. Closure shall be as specified in the appendix of the box specification.

5.1.2.5 Intermediate package. Twelve unit packages shall be packed in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

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5.1.3 Level C. Each syringe along with replacement plunger with harpoon and instructions for use, shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Twelve intermediate packages (144 unit packages) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and water-proofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Twelve intermediate packages (144 unit packages) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged syringes shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking.

5.3.1 Interior packages. Each unit and intermediate package shall be marked in accordance with MIL-STD-129. The contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The syringe covered by this specification is intended for the administration of dental local anesthetic injections.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) The National Stock Number (NSN).

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6.3 This specification is intended to cover the following item listed in the Federal Supply Catalog.

National Stock Number	Item Identification
6515-01-010-8761	SYRINGE, CARTRIDGE, ASPIRATING, THUMB RING HANDLE.

6.4 The syringe covered by this specification is suitable for use with the following items listed in the Federal Supply Catalog.

National Stock Numbers	Item Identification
6505-00-261-7240	Lidocaine Hydrochloride Injection, Modified, 2%, with Epinephrine 1:50,000, Cartridges, 1.8 ml, 50s.
6505-00-576-8842	Lidocaine Hydrochloride Injection, USP, 2.0%, with Epinephrine 1:100,000 Cartridges, 1.8 ml, 50s.
6505-00-823-7942	Mepivacaine Hydrochloride and Levonordefrin Injection, NF, 2% 1.8 ml, 50s
6505-00-982-7301	Mepivacaine Hydrochloride Injection, USP, 3%, 1.8 ml, 50s
6515-00-181-7403	Needle, Hypodermic, Cartridge Type, Disposable, 25 Gage, 13/16 inch, 100s
6515-00-181-7404	Needle, Hypodermic, Cartridge Type, Disposable, 25 Gage, 1-3/8 inches, 100s
6515-00-181-7412	Needle, Hypodermic, Cartridge Type, Disposable, 27 Gage, 1-3/8 inches, 100s
6515-00-181-7425	Needle, Hypodermic, Cartridge Type, Disposable, 27 Gage, 13/16 inch, 100s

Custodians:
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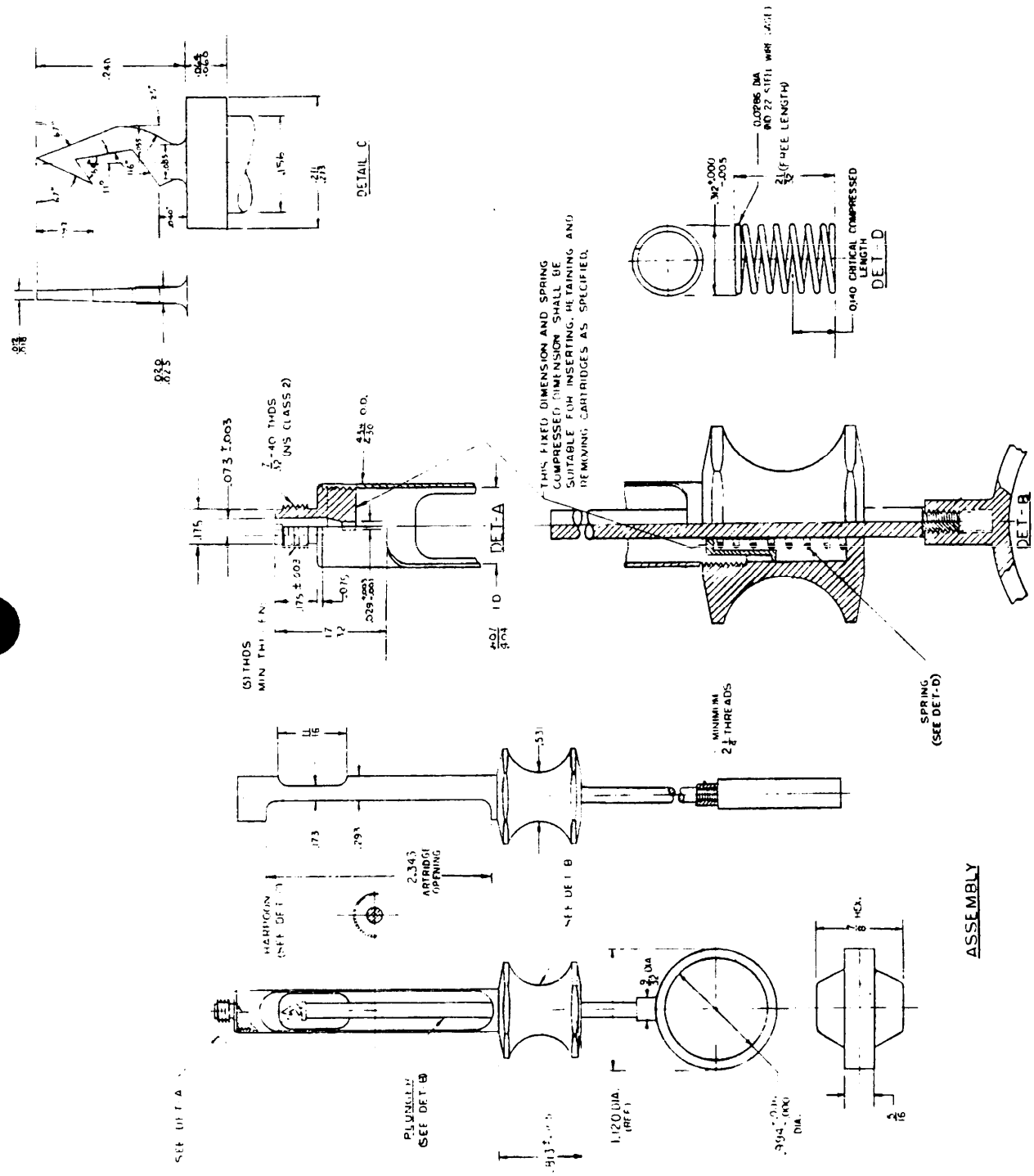


FIGURE 1. Aspirating cartridge syringe.

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DOCUMENT IDENTIFIER AND TITLE
MIL-S-37607 SYRINGE, CARTRIDGE, ASPIRATING, THUMB RING HANDLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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